
ENGINEER 3PX3

Engineering Economics

Course Outline - Winter Semester, 2024-2025



1. Course Details

Lecture Instructional Team:

Dr. Vince Leung, PhD, P.Eng.

leungv@mcmaster.ca

Dr. Matt Minnick, PhD, P.Eng.

minnick@mcmaster.ca

Instructional Assistant Intern:

Mr. Joshua Vilim

vilimj1@mcmaster.ca

Ms. Hala Zou'bi

zoubih@mcmaster.ca

Support Staff:

Mr. Basem Yassa

yassab@mcmaster.ca

Mr. Brendan Fallon

fallonbr@mcmaster.ca

Design Studio Instructional Team is all of the above plus:

Dr. Robert Fleisig, PhD, P.Eng.

robert@mcmaster.ca

Dr. Aleksandar Jeremic, PhD, P.Eng.

jeremic@mcmaster.ca

Office Hours: Available starting Week 2 to assist you with course material and address any questions you may have. They will be hosted in the "Office Hours" channel in the Engineer 3PX3 (2025) Microsoft Team.

- IAI Joshua: Thursdays, 9:00 a.m.–10:00 a.m.
- IAI Hala: Fridays, 1:00 p.m.–2:00 p.m.

Please note that the final week for office hours will be Week 12.

2. Communication Policy

Need help and have questions related to the course? We want to make sure your questions get answered. To ensure this, it's important that the correct communication method is used.

Got a question during scheduled class hours (e.g., lecture, Design Studio)? The best way to get a prompt response is to raise your hand and ask away! It is likely that others have the same question too and we can't answer a question that is not asked.

Got a question outside scheduled class hours? In this case, email is the best method. Please direct all emails to: prof3px3@mcmaster.ca. This ensures your email gets directed to the most appropriate individual for the fastest response. Every attempt will be made to reply within 24 hours (excluding weekends). Please include a subject prefix of "ENGINEER 3PX3". Emails must be sent from your @mcmaster.ca account. Be sure to include your student number in your email.

3. Inclusive Environment Statement

We consider this classroom to be a place where you will be treated with respect, and we welcome individuals of all ages, backgrounds, beliefs, ethnicities, genders, gender identities, gender expressions, national origins, religious affiliations, sexual orientations, ability – and other visible and non-visible differences. All members of

this class are expected to contribute to a respectful, welcoming, and inclusive environment for every other member of the class.

We will gladly honour your request to address you by an alternate name or gender pronoun. Please advise us of this preference early in the semester so that we may make appropriate changes to our records.

4. Course Description and Learning Outcomes

Engineering economic analysis explored in the context of an individual engineering project and an interdisciplinary engineering project. Students perform cost-benefit analysis considering the perspective of the individual and engineering team, client, and society in order to maximize net value of an engineering decision. Factors considered and incorporated include conversion factors, opportunity cost, time value of money, risk analysis, and technical analysis.

LEARNING OUTCOMES

Upon successful completion of the course, the student should be able to:	
LO.01	Utilize economic principles to make decisions in engineering projects
LO.02	Formulate Net Value functions to evaluate and compare the value and cost of alternative engineering decisions
LO.03	Make reasonable assumptions or perform necessary research to cope with ambiguity and uncertainty in required tasks
LO.04	Apply the fundamentals of cost, price, present value, and other financial metrics
LO.05	Manage group projects and interpersonal relations, especially building project management skills, preparing agendas, and diagnosing team dynamics
LO.06	Professionally communicate complex engineering decisions following an economic analysis

The Graduate Attributes defined in this section are measured for Accreditation purposes only and will not be directly taken into consideration in determining a student's actual grade in the course. For more information on Accreditation, please visit: <https://www.engineerscanada.ca>. Mapping of the Learning Outcomes outlined above to the Canadian Engineering Accreditation Board (CEAB) Graduate Attributes are outlined in the table below:

MAPPING TO GRADUATE ATTRIBUTES

GRADUATE ATTRIBUTE		LEARNING OUTCOMES
A04 Design		
4.1	Defines the problem by identifying relevant context, constraints, and prior approaches before exploring potential design solutions.	LO1, LO2
4.2	Explores a breadth of potential solutions, considering their benefits and trade-offs as they relate to the project requirements.	LO1, LO2
4.3	Develops models or prototypes, tests, evaluates, and iterates as appropriate.	LO1, LO2
A06 Individual and Teamwork		
6.1	Actively contributes to the planning and execution of a team project.	LO5
A07 Communication Skills		
7.1	Demonstrates comprehension of technical and non-technical instructions and questions.	LO6
A11 Economics and Project Management		
11.1	Applies economic principles in decision making.	LO1, LO2, LO4
11.2	Plans and effectively manages a project's time, resources, and scope, following business practices as appropriate.	LO3, LO5
11.3	Identifies, characterizes, assesses, and manages risks to project success.	LO3

5. Course Delivery

The delivery of this course occurs through weekly lectures and Design Studios. The course consists of three sections: C01, C02, and C03. Please always go to the same lecture time.

The course is run through two **50-minute** lectures and one **1 hour, 50 minute** in-person Design Studio weekly.

Lecture topics change each week and support the succeeding Design Studios. Refer to your assigned lecture section for the date and time.

ASYNCHRONOUS DESIGN STUDIO PREP represents one hour of unscheduled learning time per week (above and beyond the usual asynchronous time you would expect to spend reviewing lectures, studying, and doing assignments in a course). The activities are posted on Avenue to Learn prior to the start of each week to help you prepare for the Design Studio. The **asynchronous work for a given week must be completed before the Design Studio for that same week.**

DESIGN STUDIOS are a dedicated time for individual collaboration during Design Studio Weeks 2-3 and team collaboration during Design Studio Weeks 4-13 on engineering economic project activities. Instructional staff (i.e., faculty mentors, IAI, and TAs) will be available for support during your scheduled time.

Online Management – Avenue-to-Learn

Avenue-to-Learn will be the online management system. Through Avenue, you will be able to:

- Find all course materials (lecture slides, projects briefs, asynchronous tutorial worksheets and videos, design studio documents, etc.)
- View course-related announcements
- Submit course work (individual quizzes and assignments and group work) for grading
- View your gradebook

6. Informal Course Description

The course lectures will deliver information to help you build the skills to apply engineering economics in decision making. The Design Studios will be instructor-led and support the students in completing their engineering economic analysis and designs informing it.

The topics in this course will be taught in sections divided among lecture weeks:

- Week 1: Course Introduction, Introduction to Engineering Economics, and Net Value Introduction
- Week 2: Engineering Economics & Net Value and Detailed Conversion Factors
- Week 3: Estimation and Technical Analysis Design Implementation
- Week 4: Technical Analysis Design with Economic Impact and Sensitivity Analysis
- Week 5: Project Management and Time Value of Money
- Week 6: Time Value of Money
- Week 7: Reading Week
- Week 8: Taxes, Incentives, and Depreciation and Risk Management
- Week 9: Stochastic Net Present Value with Risks and Nonlinear Deterministic and Stochastic Models
- Week 10: Sensitivity Analysis in Stochastic Models and Confidence in Economic Analysis Involving Uncertainty
- Week 11: Supply and Demand

- Week 12: Behavioural Economics
- Week 13: Summative Synthesis and Exam Review
- Week 14: Exam Review

Design Studios consist of a brief introductory project in the first three weeks of the course that includes:

1. Consider a real-world example of given job opportunities
 - Applying topics of Conversion Factors and Cash Flow Analysis

Followed by a comprehensive project that includes:

1. Selecting a multi-disciplinary design problem and solution
 - Applying topics of Decision Making, Project Management, and Technical Analysis
2. Determine the economic viability of a project through net value functions
 - Applying topics of Net Value Functions, Conversion Factors, and Time Value of Money
3. Perform sensitivity analysis to inform project decision making
 - Applying topics of Sensitivity Analysis and Monte Carlo Simulations

7. Student Deliverables and Assessment

The following is the grade breakdown for course assessments.

- Take-Home Quizzes (8%)
 - 11 take-home quizzes; lowest three quizzes dropped.
- Individual Introductory Project Report (7%)
- Main Project Progress Check-Ins/Main Project Design Studio Participation (8%)
- Simple Report (13%)
 - Report: 9%
 - Interview: 4%
- Complex Report (26%)
 - Report: 18%
 - Interview: 8%
- Main Project Self-Reflection (3%)
- Final Exam (35%)
- Peer Evaluations:
 - Peer evaluations will be carried out after the group projects and Complex Report. The peer evaluations will be used to weigh the marks assigned to an individual group member; however, all mark weighting decisions will be at the discretion of the course co-ordinators.

Take-Home Quizzes

A take-home quiz will be released every Monday and will be due the following Monday at 11:59 p.m. for the duration of the term.

Introductory Project

Students individually will consider a scenario where they are provided with two different co-op sequences, and 3 co-op opportunities, each with different benefits and costs. Using engineering economics concepts as justification, each student will determine which sequence is a better one to pursue.

Main Project

Student teams will choose from a set of real-world design problems to take on. Project descriptions that detail the expectations will be provided. To select a project, students will need to identify where they will individually plan to use info from one or more of their past technical courses to "design" some aspect of the solution (i.e.,

use technical knowledge & analysis skill to come up with some relationship between cost and performance for some aspect of the project).

Note: you do not need to actually implement your designs by building anything or even design all aspects of the solution to work on a given project. You just need to design some aspects of it that would enable you to complete an economic analysis as detailed below.

Students will need to:

1. As a group, translate the project requirements into a formal net value function
2. As a subgroup, collaborate to "design" an aspect of the solution
3. As a group, use the results from subgroup design work to select the best combination of inputs to give the best overall solution
4. Present project progress through Check-Ins, reports, and team interviews

The individual technical work students do is not rigorously assessed; rather, as long as your work has clearly been done with due diligence on your part, we will assess your presentation of the technical work and how well you use it to complete this economic analysis.

8. Submission of Work

SUBMISSION PENALTIES: Please be aware of the following penalties for Individual Activities and Projects:

- All written reports and assignments must be uploaded to Avenue by 11:59 pm EST on the due date, or they will be subject to a late penalty.
- Submissions must be in the correct format or will be subject to a mark deduction.
- Any submissions deemed to be partially or fully copied will be considered an academic offence and be subject to terms laid out under the Academic Integrity Policy

GRADING CONCERNS: Grades will be posted to Avenue as soon as possible upon completion. You will have 7 calendar days from the date your grade is posted to address any concerns you may have to the instructional team.

- Any questions/concerns must be addressed electronically through an MS Form that will be made available on Avenue.
- Concerns with regards to grading will not be considered without submitting through the appropriate channels (i.e., the posted MS Form).
- Any concerns or appeals brought to the attention of the instructional team after 7 calendar days from the day the grade was posted to Avenue will not be processed.

9. Important Dates

Individual Introductory Project Report Due	Design Studio Week 4
Simple Report Due	Design Studio Week 8
Team Interview #1	Design Studio Week 9
Complex Report Due	Design Studio Week 12
Team Interview #2	Design Studio Week 13
Main Project Self-Reflection Due	Design Studio Week 13

Dates are subject to change. Appropriate notice will be provided. Refer to Lecture 01 – Introduction and Project Modules for more.

10. Textbooks and Course Materials

No required textbooks. All required course materials will be provided on Avenue to Learn.

11. Policy Statements

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at [University Secretariat - McMaster University Policy Directory](#).

Students are not permitted to use generative AI in this course. In alignment with [McMaster Academic Integrity Policy](#), it "shall be an offence knowingly to ... submit academic work for assessment that was purchased or acquired from another source". This includes work created by generative AI tools. Also stated in the policy is the following, "Contract Cheating is the act of "outsourcing of student work to third parties" (Lancaster & Clarke, 2016, p. 639) with or without payment." Using Generative AI tools is a form of contract cheating. Charges of academic dishonesty will be brought forward to the Office of Academic Integrity.

The following illustrates only four forms of academic dishonesty:

- Plagiarism, e.g. submission of work not one's own or which other credit been obtained.
- Improper collaboration in group work.
- Copying or using unauthorized aids in tests and examinations.
- Using generative AI to complete quizzes or assist with report writing.

AUTHENTICITY / PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. In those courses, students will be expected to submit their work electronically either directly to Turnitin.com or via Avenue to Learn (A2L) plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish to submit their work through A2L and/or Turnitin.com must still submit an electronic and/or hardcopy to the instructor. No penalty will be assigned to a student who does not submit work to Turnitin.com or A2L. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). To see the Turnitin.com Policy, please go to www.mcmaster.ca/academicintegrity.

ON-LINE ACCESS

In this course, we will be using **Avenue-to-Learn** and **Microsoft Teams**. Students should be aware that, when they access the electronic components of this course, private information such as first and last names, usernames for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure, please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

ACADEMIC ACCOMMODATIONS FOR STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy.

ACADEMIC ACCOMMODATION FOR RELIEF FOR MISSED ACADEMIC WORK – McMaster Student Absence Form (MSAF)

In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work"

1. All MSAFs are to be directed to prof3px3@mcmaster.ca. Sending to another email address will delay processing.
2. It is the prerogative of the instructor of the course to determine the appropriate relief for missed term work in his/her course.
 - For **individual assignments**, the grade weight of the MSAFed assessment will be added to the weight of the Final Exam. MSAFed project **interviews** will be rescheduled to a later date.
 - For individual MSAFs used on **group submissions**, the group must submit what they have at the time of the original deadline and re-submit if granted an extension on a case-by-case basis.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.

NOTICE REGARDING POSSIBLE COURSE MODIFICATION

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

REFERENCE TO RESEARCH ETHICS

The two principles underlying integrity in research in a university setting are these: a researcher must be honest in proposing, seeking support for, conducting, and reporting research; a researcher must respect the rights of others in these activities. Any departure from these principles will diminish the integrity of the research enterprise. This policy applies to all those conducting research at or under the aegis of McMaster University. It is incumbent upon all members of the university community to practice and to promote ethical behaviour. To see the Policy on Research Ethics at McMaster University, please go to <https://reo.mcmaster.ca/>.